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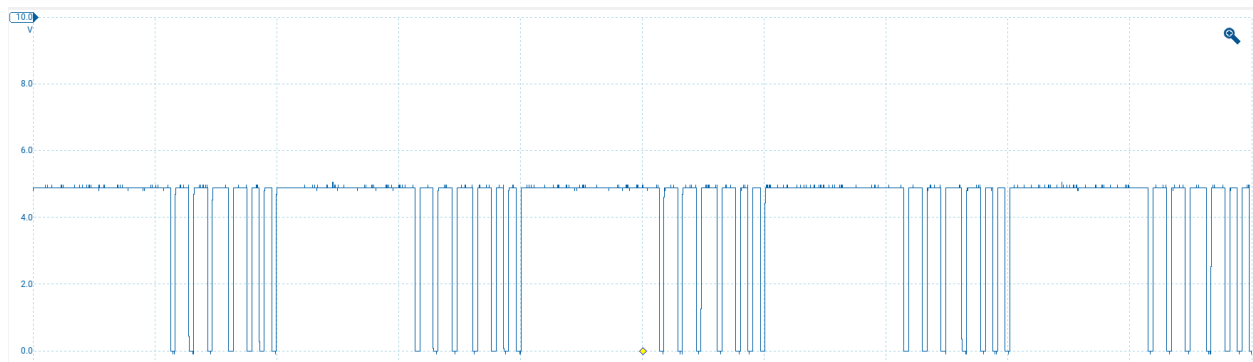
MP-1 Lab Report

In this lab we were tasked to design a hardware/software system to capture/generate and modify ppm data frame Hobby King transmitter and eventually control a small quadcopter. This project involved heavy design/development using Xilinx development tools, FPGA programming, VHDL, and embedded C.

PRACTICE WITH THE QUAD:

When flying the drone we found the throttle to be the most sensitive aspect of flying. Its controls are also in reference to the “head” of the drone so you must remember where your heading is. We kept messing with the trim levels in an attempt to get the drone to mostly stay still while ascending, with some success. It wasn’t ever *perfect*, but it was OK.

PPM SIGNALS:



This is a 6-channel PPM, with a frame width of 20ms, widest observed pulses are ~1.5ms at and shortest observed pulses ~0.5ms, with an inter-channel gap of 400us. From subtracting the total time of all 6 channels (with gaps) from the frame width (20ms) a minimum idle pulse of ~8.2ms.

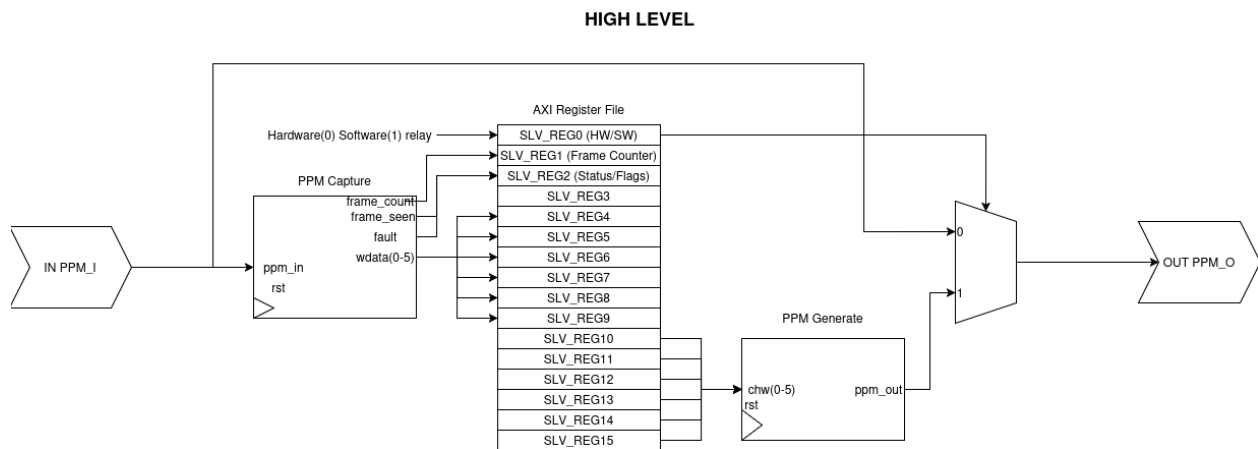
$$20ms(\text{framewidth}) - ((7*0.4) + (6*1.5)) = \sim 8.2ms$$

Pulse 1	Pulse 2	Pulse 3	Pulse 4	Pulse 5	Pulse 6
Right stick Horizontal	Right stick Vertical	Left Stick Vertical	Left Stick Horizontal	Right Knob	Left Knob

CONCERNS:

The output of the trainer port is at 5v. while the port on the FPGA runs on 3.3v, to rectify this issue we are using a level shifter to take our 5v signal down to 3.3v. We found this specification within the .xdc file for the ZedBoard, and we already had to use the level shifters when we were trying to get the joystick working for MP-0.

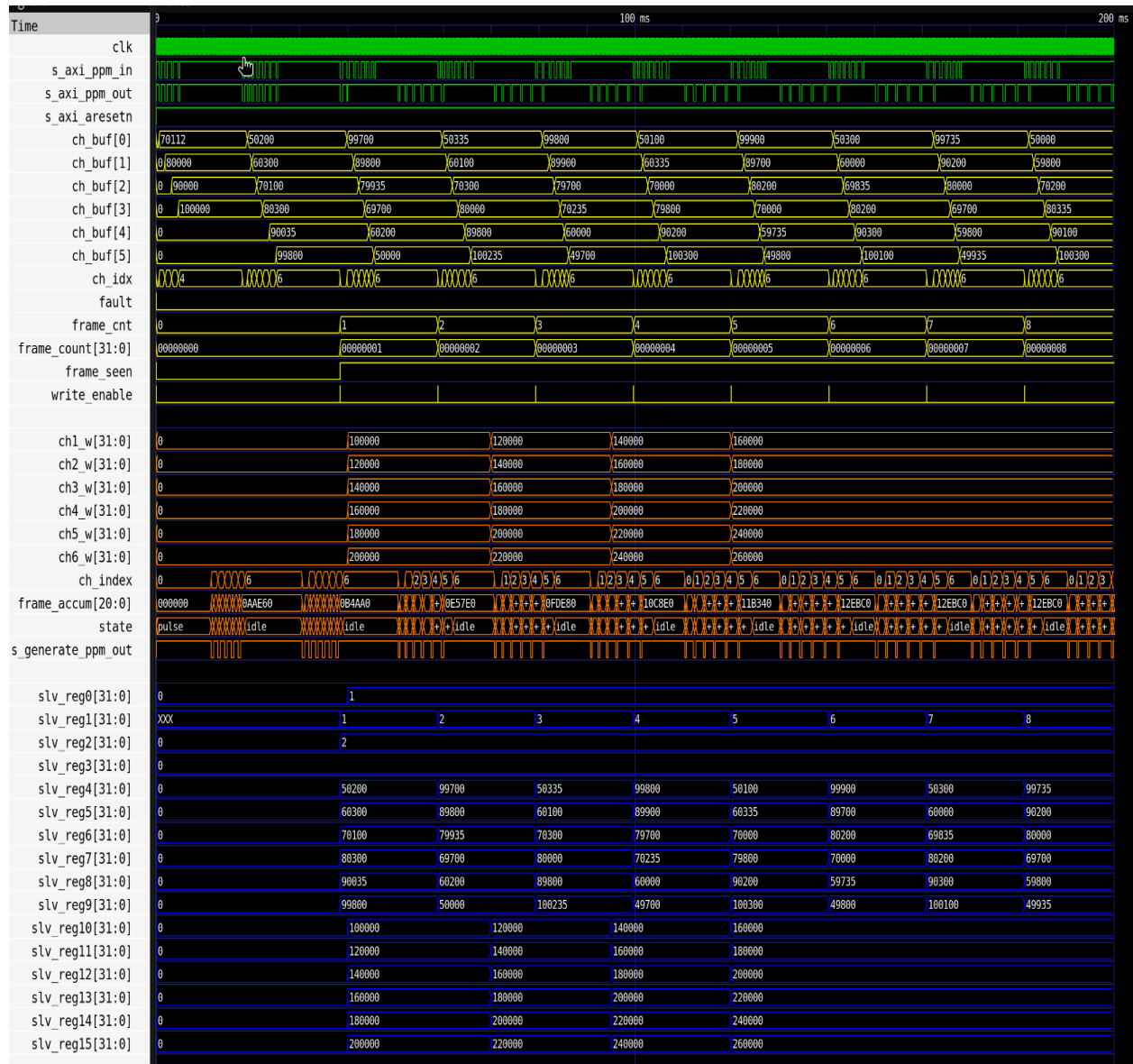
HARDWARE IMPLEMENTATION:



High-Level Design Schematic of out PPM

When the AXI master provides an address on the bus, the slave first latches it and checks it against its register map. A write enable (slv_reg_wren) is generated only when both the write address and write data are valid and match a register. A read enable (slv_reg_rden) is generated when a valid read address is accepted, allowing the corresponding register's data to be placed on RDATA. The address decoding ensures only the targeted register responds.

The PPM state machine directly connects to the memory-mapped registers inside the AXI slave logic. For capture, it writes pulse widths and status flags to the registers; for generation, it reads channel widths from the registers. All accesses are synchronous to the system clock, so the state machine and AXI interface safely share the registers.



Waveform from running capture(yellow),generate(orange) and registers(blue) in simulation

PPM CAPTURE

The ppm_capture module captures a PPM signal from an RC receiver by synchronizing and filtering the input, then measuring the width of each high pulse. On each falling edge, it stores the pulse width as a channel value or detects a long high pulse as a frame sync. When a full frame of channels is collected, it latches the values, pulses write_enable for one cycle, updates the frame count, and tracks signal faults. Outputs wdata1..6 hold the latest channel pulse widths.

PPM GENERATE

The ppm_generate module generates a PPM signal by sequencing six channel pulses with configurable widths. It cycles through three states: IDLE , GAP , and PULSE. On each clock, it counts down a timer, outputs ppm_out high or low depending on the state, and updates the next pulse or idle duration to keep a consistent frame length of ~20 ms. Channel widths come from ch1_w..ch6_w, defaulting to 1 ms if zero.

SOFTWARE

The C software running rc_control.c reads slv_reg4-9 and performs a series of operations to them then writes them back into slv_reg10-15, those registers are used by ppm_generate to output them back. Rc_control.c also provides user interface/diagnostic information.

RECORD/PLAYBACK

The switches on the FPGA control between recording/playback modes. In recording mode all Captured signals are recorded into a frame_buffer. In playback mode the PPM signals are transmitted back to the drone, the buttons on the FPGA can be used to forward/rewind during playback.

SOFTWARE FILTERING

PPM signals captured are put through a filtering process, first values beyond a certain min/max are clamped, values are then smoothed using exponential smoothing.

USER INTERFACE

We have created a serial-interface which shows the current state of the RC controller. As well as the active mode. An optional debugging interface (by activating SW2) displays the current values inside slv_reg4-9(capture) and slv_reg10-15(generate).

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=====
RC CONTROL
=====
SOFTWARE RELAY MODE ACTIVE
Record Mode: OFF | Elapsed: 17.72 s0 s
Playback Mode: OFF | Elapsed: 2.22 s
Software Filtering ACTIVE

          .....
          .....
          .....
          .....
          .....
          .....O.....
          .....
          .....
          .....
          .....O.....
          .....
L: [----O----]  R: [----O----]
=====
```

Image of RCCONTROL user-interface

